Earth and Planetary Science Letters, 73 (1985) 140–146
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

[4]

Paleomagnetic fold test evidence for Laramide age of monoclinal folding, Salt River Canyon, Arizona

Stephen A. Sandberg and Robert F. Butler

Department of Geosciences, University of Arizona, Tucson, AZ 85721 (U.S.A.)

Received June 12, 1984
Revised version received November 27, 1984

A 1.15 b.y. old diabase sill is exposed in the Rock Canyon monocline of the Salt River Canyon within the Transition Zone physiographic province of east-central Arizona. Various interpretations have been advanced regarding the age relationship of monoclinal folding and sill intrusion. Seventy oriented samples for paleomagnetic analysis were collected from 10 sites within the diabase sill through the hinge zone of the Rock Creek monocline. Alternating field demagnetization to peak fields of 50–70 mT was sufficient to isolate the characteristic natural remanent magnetization, which is a thermoremanent magnetism acquired during cooling of the sill. Statistical analysis of the resulting paleomagnetic data indicates passage of the fold test at the 95% confidence level. The monoclinal folding clearly post-dates intrusion of the diabase sill. When added to other geologic evidence, these data indicate that the Rock Canyon monocline is a Laramide compressional feature.

1. Introduction

Various interpretations have been advanced regarding the age of monoclines located in the Salt River Canyon area of east-central Arizona. Shride [1] states that the monoclines were probably late Proterozoic features formed before the intrusion of the numerous 1.15 b.y. old diabase sills of the region [2]. Granger and Raup [3] linked the monoclines of this area with steeply dipping Precambrian fault zones at depth and concluded that these faults served as conduits for emplacement of the diabase sills. Thus, Granger and Raup [3] state that "the monoclines clearly were formed before or during diabase intrusion as is indicated by their great influence on the emplacement of the diabase masses." However, Davis et al. [4] interpreted the Salt River Canyon monoclines to be Laramide compressional features similar to the monoclines of the Colorado Plateau. If the Laramide age is correct, the monoclines of the Salt River Canyon would have formed over 1000 m.y. after diabase intrusion.

A diabase sill is exposed in the Rock Canyon monocline (Fig. 1) near the level of the Salt River. Davis et al. [4] postulated that the excellent exposure of the hinge zone of this monocline might provide the opportunity for determination of the relative timing of intrusion of the diabase and flexure of the monocline using paleomagnetic techniques. If intrusion of the diabase sill produced the monoclinal flexure, the paleomagnetism should fail the fold test of Graham [5] because the characteristic natural remanent magnetism (NRM) would have been acquired during cooling of the sill after formation of the monocline. In this case, the characteristic NRM directions from sampling sites through the monoclinal flexure should be well grouped in situ but should diverge upon structural correction. However, if the monocline is, in fact, a Laramide compressional feature, the paleomagnetism of the diabase should pass the fold test. In that case, the characteristic NRM directions should be divergent in situ but should converge upon structural correction. Helsley [6] and Helsley and Spall [7] demonstrated that the diabase sills in the

0012-821X/85/$03.30 © 1985 Elsevier Science Publishers B.V.

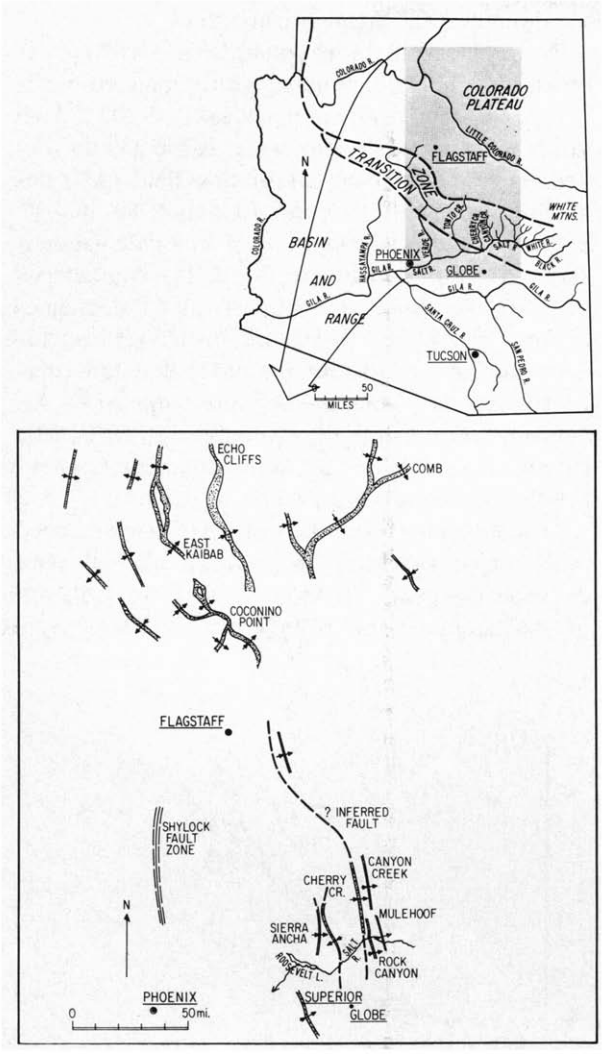


Fig. 1. Locations of monoclines in the Salt River Canyon area [3] and locations of monoclines in northern Arizona section of Colorado Plateau [8] (after Davis et al. [4]). Insert shows physiographic setting.

Salt River Canyon and Sierra Ancha region possess a stable natural remanent magnetization. Results of a paleomagnetic fold test on the diabase sill exposed in the Rock Canyon monocline are presented here.

2. Geologic setting

The Rock Canyon monocline is one of five monoclines exposed in the Salt River Canyon area. As shown in Fig. 1, the Salt River monoclines are located in the Transition Zone physiographic province. Also shown are the locations of nearby monoclines on the Colorado Plateau [8].

The Rock Canyon monocline is an east-facing fold which can be traced some 7 km and is the only monocline in the Salt River Canyon for which both hinge zones are exposed [4]. The monocline displays a minimum of 90 m of structural relief, trends about N22°W, and locally plunges about 10° towards S20°E [3,4]. Upper Proterozoic Apache Group sedimentary rocks, including the Mescal Limestone and the Dripping Spring Quartzite, are deformed by the Rock Creek monocline. The diabase sill exposed in the monocline intrudes the Dripping Spring Quartzite and is just one element of an extensive complex of sills that have intruded Apache Group sedimentary rocks in the Salt River Canyon area. Thickness of the diabase sill in the Rock Canyon monocline is about 100 m.

Silver [9] reported an average age of approximately 1120 ± 10 m.y. from a number of U/Pb dates on cogenetic zircons within granophyric differentiates of diabase sills in the Sierra Ancha area. Analysis by use of a concordia diagram yielded an age determination of 1150 ± 30 m.y. [2]. Damon et al. [10] reported a K/Ar date of 1140 ± 40 m.y. for biotite from the Sierra Ancha diabase.

3. Paleomagnetism

Seventy oriented core samples were collected from 10 sites in the diabase sill of the Rock Canyon monocline exposed on the north side of the Salt River Canyon. Six to eight oriented cores were collected at each site using standard techniques similar to those described by Doell and Cox [11]. Cores were oriented with Brunton compass. Seven sites were collected from the steeply dipping limb where the contact between the diabase and the overlying Dripping Spring Quartzite displays dips ranging from 60 to 80°. Three sites were collected west of the anticlinal hinge where dips are approximately 20°.

Lack of adequate exposure and zones of penetrative fracturing within the diabase limited

the number of acceptable sites for paleomagnetic sampling. Sites were placed such that the structural attitude could be reasonably well determined. For sites within 10 m of the contact with the Dripping Spring Quartzite, the structural attitude of bedding planes within the quartzite were used. A bed of approximately 2 m thickness of Dripping Spring Quartzite is incorporated within the sill. This bed extends for approximately 35 m, is parallel to the contact between the diabase and the overlying Dripping Spring Quartzite, and is about 20 m below that contact. Isolated beds of sedimentary rock within diabase sills are common in the region and most parallel bedding within the sedimentary rocks above and below the sill [1]. It is believed that attitudes of bedding surfaces within this layer of quartzite closely approximate the structural attitude of the diabase within about 10 m of the layer. Extrapolation of structural attitudes, even over short distances, is likely to be subject to error due to local deformation in the hinge zone of the fold.

Strong-field thermomagnetic experiments were performed on representative samples from 3 sites. A typical result is illustrated in Fig. 2. All 3 samples produced reversible heating and cooling curves and a single Curie temperature at 580°C. This Curie temperature indicates that magnetite is the dominant ferrimagnetic mineral.

Natural remanent magnetization (NRM) was measured using a flux-gate spinner magnetometer or a cryogenic magnetometer (ScT C-102). Two samples from each site were subsequently examined by progressive alternating field (AF) demagnetization up to peak fields of 80 mT (1 mT = 10 Oe). A representative vector demagnetization diagram is shown in Fig. 3. The characteristic NRM was isolated in all cases at AF demagnetization levels between 50 and 70 mT. The vector demagnetization diagrams indicate that low-coercivity secondary components were removed by AF demagnetization to approximately 50 mT. The characteristic NRM is a thermoremanent magnetization acquired during cooling of the sill.

All remaining samples were AF demagnetized in several peak fields. These demagnetization levels spanned the peak AF level required for isolation of the characteristic NRM in the progressively

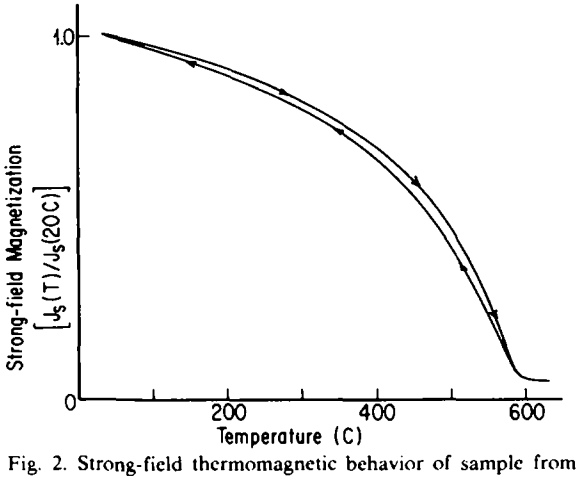


Fig. 2. Strong-field thermomagnetic behavior of sample from site SR003. Experiment was performed in 1/2 atmosphere argon gas with heating and cooling rate of 20°C/minute. Arrows indicate heating and cooling curves.

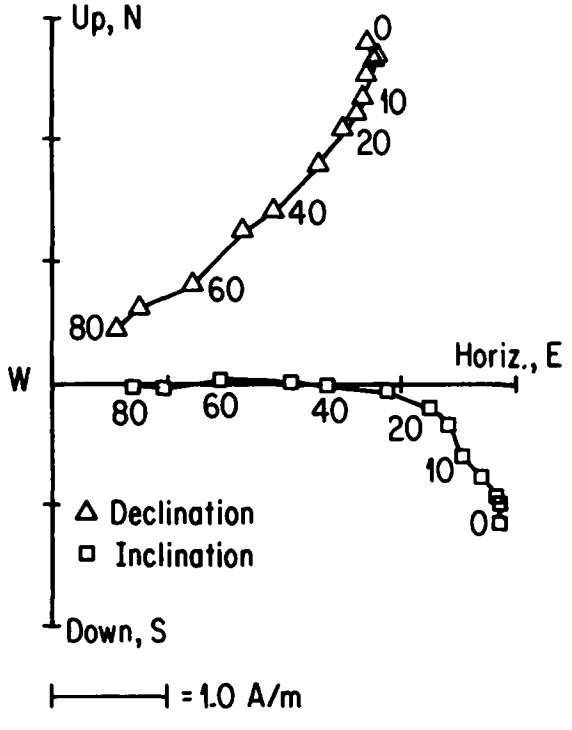


Fig. 3. Vector demagnetization diagram illustrating typical response to progressive AF demagnetization. Numbers adjacent to data points indicate AF demagnetization treatment in mT.

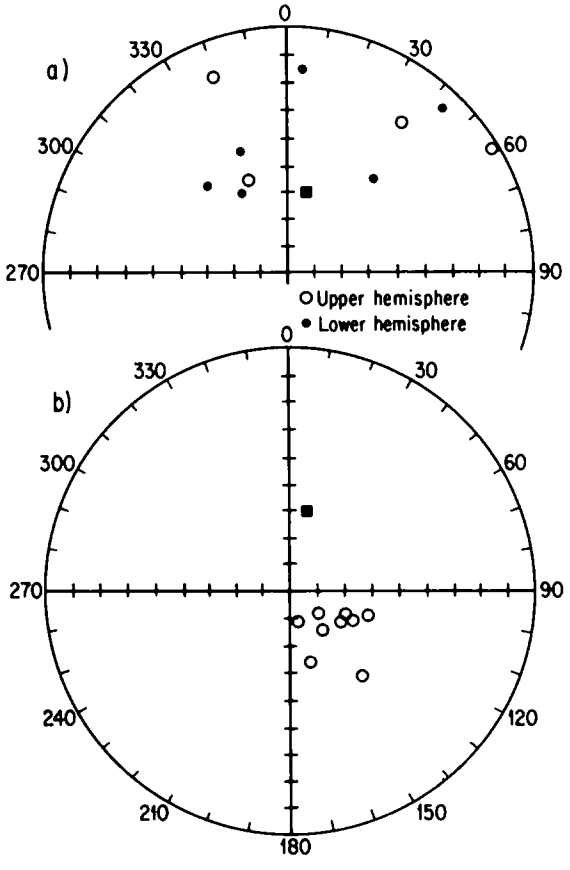


Fig. 4. (a) Equal area stereographic projection of initial NRM directions for samples from site SR003 before AF demagnetization. (b) Characteristic NRM direction for site SR003 after AF demagnetization to 60 mT.

demagnetized samples from that site. The optimal demagnetization treatment for each site was chosen primarily as the level of demagnetization which was within the range of AF treatments which produced a linear trend toward the origin on vector demagnetization plots. The site mean directions of characteristic NRM were stable over a significant range of AF treatments. Stereographic projections of NRM directions for a representative site before and after AF demagnetization are shown in Fig. 4. Site mean results, along with the relevant Fisher [12] statistical parameters, are listed in Table 1.

4. Paleomagnetic fold test

The classic fold test of Graham [5] was used to determine the relative timing between acquisition of the characteristic NRM by the diabase during cooling and the flexure of the Rock Creek monocline. The 10 site mean directions were structurally corrected by removal of the 10° plunge toward S20°E followed by tectonic correction utilizing the structural attitude measured near the site. Fig. 5 illustrates the site mean directions before (uncorrected) and after (corrected) structural correction. It is evident that the site mean directions converge following structural correction.

One technique for evaluation of the statistical significance of the fold test is to employ the proce-

TABLE 1

Summary of site mean results

Site	AF level (mT)	Uncorrected		Corrected		Attitude (°)	J (A/m)	α_{95} (°)	k	R	N
		I(°)	D(°)	I(°)	D(°)						
SR002	70	−28.4	72.9	−73.3	301.6	170, 74	1.0×10^{-1}	14.4	18.5	6.67	7
SR003	60	−68.0	132.6	−71.9	198.3	148, 22	5.6×10^{-2}	8.6	42.5	7.84	8
SR004	60	−69.8	101.9	−82.9	199.0	148, 22	7.1×10^{-2}	9.0	56.5	5.91	6
SR005	60	−74.2	150.2	−68.6	221.3	148, 22	7.6×10^{-2}	9.6	50.2	5.90	6
SR006	50	4.5	38.2	−44.5	15.6	170, 79	4.8×10^{-1}	4.2	259.2	5.98	6
SR007	50	−20.0	68.4	−74.5	330.9	170, 75	1.3×10^{-1}	8.2	68.5	5.93	6
SR008	60	−49.0	106.4	−63.3	239.7	170, 60	2.9	5.3	129.3	6.95	7
SR009	50	8.9	45.3	−43.8	32.7	170, 70	6.0×10^{-1}	4.3	198.8	6.97	7
SR010	60	−2.1	51.6	−55.0	23.6	170, 70	9.6×10^{-1}	4.1	262.8	5.98	6
SR011	60	−12.7	45.9	−53.5	3.2	170, 70	1.0	5.3	212.7	4.98	5

AF level is peak alternating field used in demagnetization treatment. I and D are site mean inclination and declination respectively. J is intensity of remanent magnetization following AF demagnetization. α_{95}, k, R are statistical parameters of Fisher [12]. N is number of samples. First number of the attitude is azimuth 90° clockwise from down dip, while second number is dip.

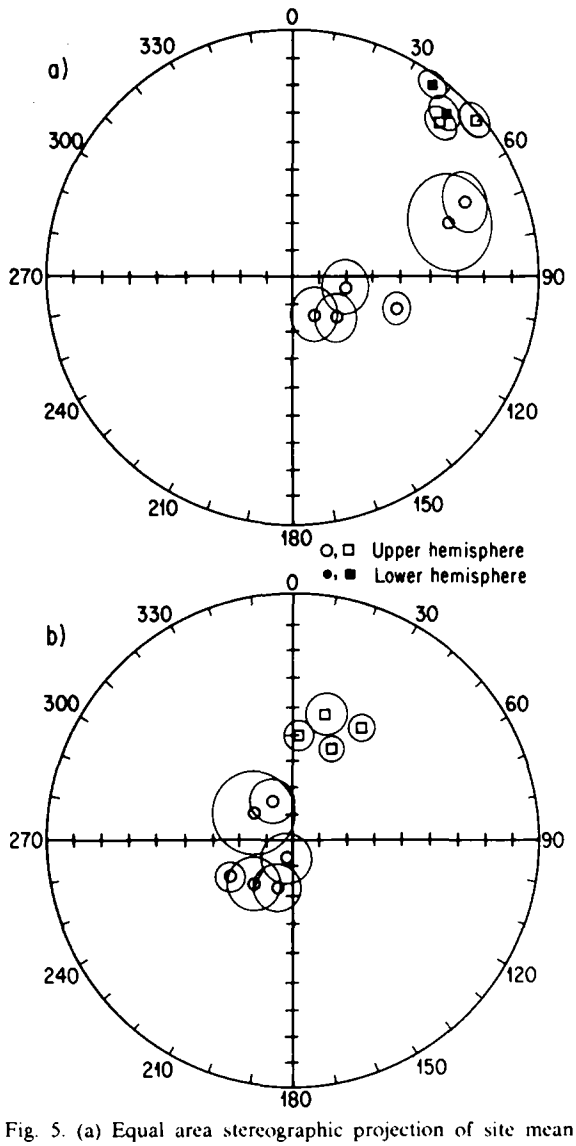


Fig. 5. (a) Equal area stereographic projection of site mean directions of characteristic NRM before structural correction. (b) Site mean characteristic NRM directions after structural correction. Oval of 95% confidence is shown for each site mean. Squares indicate characteristic NRM directions for sites near upper contact of sill.

dures of Watson [13] and Watson and Irving [14] as tabulated by McElhinny [15]. This procedure compares the best estimate of the Fisher [12] precision parameter (k) for the corrected site mean directions to the precision of the uncorrected site mean directions. For the data set from the 10 site mean directions in the diabase sill, the ratio of k_2 (corrected) to k_1 (uncorrected) is 2.00. A fold test with $k_2/k_1 > 2.22$ (for $N = 10$) is significant at the 95% confidence level [15], while $k_2/k_1 > 1.86$ indicates significance at the 90% confidence level [15]. On the basis of this technique, we can state with 90% confidence that the characteristic NRM of the diabase was acquired before flexure of the Rock Canyon monocline. However, as described below, some inadequacies are evident in the above fold test on the 10 site mean directions.

The structurally corrected site mean vectors (Fig. 5b) show two groupings of directions. One grouping is from sites within the interior of the sill about 20 m below the upper contact, while the other grouping is from sites collected on the steeply dipping limb near the contact with the overlying Dripping Spring Quartzite. The mean directions for these two groups of directions are, in fact, statistically distinguishable at the 95% confidence level. Because of secular variation during cooling of the sill, it is quite likely that the characteristic NRM from sites near the upper contact recorded a different geomagnetic field direction than that recorded by the interior sites. A calculation of conductive cooling for a 100 m thick sill [16] indicates that the upper contact would cool to 600°C almost immediately upon intrusion. At least 100 years would be required for the 600°C isotherm to pass through a point 20 m below the upper contact. This is certainly sufficient time for significant geomagnetic secular variation to have occurred between magnetization of sites adjacent to the upper contact and sites which are 20 m below that contact.

Although it was not logistically possible to do so, collection of sites in the mildly dipping limb near the upper contact would have permitted separate fold tests to be performed on sites interior to the sill and on sites near the upper contact. However, with the available data it is most appropriate to apply the fold test only to the sites from within the interior portion of the sill.

There are two approaches to statistical analysis of the fold test for the six interior sites. The statistical significance of the fold test on these interior sites can be evaluated by the method of Watson [13] and Watson and Irving [14] (Table 2).

TABLE 2

Fold test parameters for $N = 6$ interior data set

Sites 2, 7, 8	$R_a = 2.921$
Sites 3, 4, 5	$R_b = 2.979$
Overall:	
uncorrected	$R_1 = 5.377$ $k_1 = 8.03$
corrected	$R_2 = 5.828$ $k_2 = 29.03$

R and k are statistical parameters of Fisher [12].

Using these methods, k_2/k_1 is 3.62, a value which is greater than 2.97, the variance ratio of the F-distribution at the 95% confidence level for $N = 6$ [15]. Thus the data set from the 6 interior sites easily passes the fold test at the 95% confidence level.

An alternative approach to statistical evaluation of the fold test has been developed by McFadden and Jones [17]. They argue that the significance of the fold test should be based on "whether the mean direction of a group of sites from one limb of a fold may be distinguished statistically from the mean direction of a group of sites from another limb". The assumption that two limbs of a fold share a common true mean direction may be rejected if:

$$\frac{(R_a + R_b - R^2/(R_a + R_b))}{2(N - R_a - R_b)} > (1/p)^{1/(N-2)} - 1 \quad (1)$$

where R_a = resultant length of the site mean unit vectors from one limb of the fold, R_b = resultant length from other limb, R = overall resultant length (either corrected or uncorrected), N = number of site mean directions, and p = level of significance.

For the $N = 6$ data set from sites interior to the diabase sill, the hypothesis of a common true mean direction may be rejected at the 95% confidence level if:

$$\frac{5.900 - R^2/5.900}{0.200} > 1.115 \quad (2)$$

where $1.115 = (1/p)^{1/(N-2)} - 1$ for $N = 6$ and $p = 0.05$. Setting $R = R_1$ (uncorrected), the left side of (2) equals 4.989 and the hypothesis of a common true mean before structural correction may be rejected. Setting $R = R_2$ (corrected), the left side of (2) equals 0.716 and the hypothesis of a common mean direction after structural correction may not be rejected. Thus, the data set from the 6 interior sites passes the fold test of McFadden and Jones [17] at the 95% confidence level.

Regardless of the method of statistical analysis, it is clear that the characteristic NRM of the diabase sill passes the fold test at a high level of confidence. The fundamental result of the paleomagnetic analysis is that the characteristic NRM of the diabase sill was acquired before flexure of the Rock Creek monocline.

5. Discussion and conclusions

Helsley and Spall [7] determined a paleomagnetic pole position (27°N, 179°E) from a number of normal polarity sites in the 1.15 b.y. old diabase sills of the Globe–Salt River Canyon area to the east and southeast of the Rock Canyon monocline. This pole position is in close agreement with a virtual geomagnetic pole determined from diabase in the Sierra Ancha area, west of our location [6]. Helsley and Spall [7] did not observe any reversed polarity sites among the Precambrian diabase sills which they sampled in central Arizona. However, the diabase sill exposed in the Rock Canyon monocline has reversed polarity. This observation indicates that sill intrusion took place during at least two intervals of opposite polarity of the geomagnetic field.

The fold test analysis indicates that the characteristic NRM of the diabase sill was acquired before the Rock Canyon monocline was folded. The monoclinal flexure thus cannot have been formed either before or during diabase intrusion. Davis et al. [4] have summarized the evidence for a Laramide age of the Salt River Canyon monoclines. Four of the five monoclines of this area face east and all trend north-northwest. These characteristics are common to monoclines of the Colorado Plateau in northern Arizona which are clearly post-Cretaceous and probably Laramide in age [18,19]. The near vertical dips of the Rock Canyon monocline in conjunction with its deep

level of exposure agrees with the similarly exposed Palisades and East Kaibab monoclines of the Grand Canyon [20] as well as the Hunters Point monocline of northeastern Arizona [19]. Regional strain analysis indicates that no Paleozoic or pre-Laramide Mesozoic deformations capable of producing the Rock Canyon monocline have affected this region [21].

The paleomagnetic fold test indicates that the monoclines of the Salt River Canyon were produced after intrusion of the diabase sills. The additional geological data indicate that a Laramide age is most probable.

Acknowledgements

We thank Gary Calderone, Scott Showalter, Gayle Weiss, and Kathy Flanagan for assistance in the field. We thank George Davis for suggesting this project and for helpful discussions. The manuscript was improved by critical readings by George Davis and Randy Richardson. Funding was provided by NSF grants EAR 8024274 and EAR 8115430.

References

1 A.F. Shride, Younger Precambrian geology in southern Arizona, U.S. Geol. Surv. Prof. Pap. 566, 1–89, 1967.
2 L.T. Silver, The use of cogenetic uranium-lead isotope systems in zircons in geochronology, in: Radioactive Dating, p. 279, International Atomic Energy Egency, Vienna, 1963.
3 H.C. Granger and R.B. Raup, Geology of uranium deposits in the Dripping Spring Quartzite, Gila County, Arizona, U.S. Geol. Surv. Prof. Pap. 595, 1–108, 1969.
4 G.H. Davis, S.R. Showalter, G.S. Benson, L.S. McCalmont and F.W. Cropp, Guide to the geology of the Salt River Canyon region, Arizona, Ariz. Geol. Soc. Digest XIII, 47–98, 1981.
5 J.W. Graham, The stability and significance of magnetism in sedimentary rocks, J. Geophys. Res. 54, 131–167, 1949.
6 C.E. Helsley, Paleomagnetic results from Precambrian rocks of central Arizona and Duluth, Minnesota (abstract), EOS Trans. Am. Geophys. Union 46, 67, 1965.
7 C.E. Helsley and H. Spall, Paleomagnetism of 1140 to 1150 million-year diabase sills from Gila County, Arizona, J. Geophys. Res. 77, 2115–2128, 1972.
8 G.H. Davis and C.W. Kiven, Structure map of folds in Phanerozoic rocks, Colorado Plateau tectonic province of Arizona, Arizona Oil and Gas Conservation Commission, Phoenix, Ariz., 1975.
9 L.T. Silver, Precambrian formations and Precambrian history in Cochise County, southeastern Arizona, in: Land of Cochise, J.G. Callender, J.C. Wilt and R.E. Clemons, eds., N.M. Geol. Soc. 29th Field Conf., pp. 157–163, 1978.
10 P.E. Damon, D.E. Livingston and R.C. Erickson, New K-Ar dates for the Precambrian of Pinal, Gila, Yavapai and Coconino Counties, Arizona, N.M. Geol. Soc. Field Conf. Guidebook 13, 56–57, 1962.
11 R.R. Doell and A. Cox, Paleomagnetic sampling with a portable coring drill, in: Methods in Paleomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds., pp. 21–25, Elsevier, Amsterdam, 1967.
12 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London, Ser. A. 217, 295–305, 1953.
13 G.S. Watson, Analysis of dispersion on a sphere, Mon. Not. R. Astron. Soc. London, Geophys. Suppl. 7, 153–159, 1956.
14 G.S. Watson and E. Irving, Statistical methods in rock magnetism, Mon. Not. R. Astron. Soc. London, Geophys. Suppl. 7, 289–300, 1957.
15 M.W. McElhinny, Statistical significance of fold test in paleomagnetism, Geophys. J.R. Astron Soc. 8, 338–340, 1964.
16 J.C. Jaeger, Cooling and solidification of igneous rocks, in: Basalts, Vol II, H.H. Hess, ed., pp. 503–536, J. Wiley and Sons, New York, N.Y., 1968.
17 P.L. McFadden and D.L. Jones, The fold test in paleomagnetism, Geophys. J.R. Astron. Soc. 67, 53–58, 1981.
18 V.C. Kelley, Monoclines of the Colorado Plateau, Geol. Soc. Am. Bull. 66, 789–804, 1955.
19 G.H. Davis, The monocline fold pattern of the Colorado Plateau, in: Laramide Folding Associated with Basement Block Faulting in the Western United States, V. Matthews III, ed., Geol. Soc. Am. Mem. 151, 215–234, 1978.
20 Z. Reches, Development of monoclines, Part I. Structure of the Palisades Creek branch of the East Kaibab monocline, Grand Canyon, Arizona, in: Laramide Folding Associated with Basement Block Faulting in the Western United States, V. Matthews III, ed., Geol. Soc. Am. Mem. 151, 235–272, 1978.
21 G.H. Davis, Regional strain analysis of the superposed deformations in southeastern Arizona and the eastern Great Basin, in: Relations of Tectonics to Ore Deposits in the Southern Cordillera, W.R. Dickinson and W.D. Payne, eds., Ariz. Geol. Soc. Digest XIV, 155–172, 1981.